Review of Economics and Statistics Just Accepted MS.
doi:10.1162/REST_a_00703
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Shops and the City

Evidence on Local Externalities and Local Government Policy from Big-Box Bankruptcies*

Daniel Shoag† and Stan Veuger‡

May 22, 2017

Abstract

*We are grateful to Daron Acemoglu, Marianne Bertrand, George Borjas, Leah Brooks, Denise DiPasquale, Edward Glaeser, Adam Guren, Christopher Palmer, Michael Strain, Matthew Turner, Daniel Wilson, Kristin Wilson, and Owen Zidar for their valuable help and suggestions. We also thank seminar participants at the Annual Conference of the Eastern Economic Association, the Annual Conference of the Southern Economic Association, Bar-Ilan University, Ben-Gurion University, the Berkeley Haas School of Business, the Duke-ERID Conference on Advances and Applications of Spatial Equilibrium in Economics, the Harvard Department of Economics, the Harvard Kennedy School, the Hebrew University of Jerusalem, the National Tax Association's Annual Conference on Taxation, the NBER Summer Institute, the North American Meetings of the Regional Science Association International, Tel Aviv University, and the World Congress of the Econometric Society for their insightful comments. Nikolai Boboshko, Nicholas Carollo, Hao-Kai Pai, Brittany Pineros, and Cody Tuttle provided excellent research assistance, while Stacey Bogan provided very helpful geographic analysis support.

†Harvard Kennedy School, 79 John F. Kennedy Street, Cambridge, MA 02138, United States, dan_shoag@hks.harvard.edu, (617) 495-7649

‡American Enterprise Institute, 1789 Massachusetts Avenue NW, Washington, DC 20036, United States, stan.veuger@aei.org, (202) 862-5894

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doi:10.1162/REST_a_00703
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We report three findings: (1) Using evidence from chain bankruptcies and data on 12-18 million establishments per year, we show that large retailers produce significant positive spillovers. (2) Local governments respond to the size of these externalities. When a town's boundaries allow it to capture a larger share of retail spillovers, it is more likely to offer retail subsidies. (3) These subsidies partially crowd out private-sector mechanisms that also subsidize large retailers, such as shopping malls. These facts provide powerful evidence of the Coase theorem at work and highlight a concern for local development policies even when externalities can be targeted.

JEL: H2, H4, H7, L5, L8, O1, R1

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I Local Governments and Retail Externalities

The Concise Encyclopedia of Economics begins its article on public goods and externalities (Cowen, 1993) with the statement that "[m]ost economic arguments for government intervention are based on the idea that the marketplace cannot... handle externalities." The article explains that "markets often [do] solve (..) externalities problems" when property rights are well defined (as in Coase, 1960). This idea, that there are often both public and private mechanisms for internalizing externalities, has been central to public finance but difficult for empirical researchers to observe in all of its implications. There are few situations where an externality can be identified in a dataset, fewer where the relative size of this externality can be correlated with policy decisions, and fewer still where these decisions can be explored alongside alternative private internalization systems.

In this paper, we study precisely such externalities, policies, and private mechanisms.[1] We set out by estimating the size of a specific type of externalities: the spillovers produced by big-box retail stores (see Basker et al., 2012, for an analysis of the rise of big-box retail chains). The question of whether and how much they affect surrounding businesses is a contentious one that is of interest to economists and policymakers alike. To estimate the size of these externalities, we exploit the nation-wide store closings associated with the bankruptcies of national chains of bookstores (Borders), department stores (Mervyn's), electronics stores (Circuit City Stores and CompUSA), and housewares stores (Linens 'n Things). We contrast the sudden, locally exogenous disappearance of these chain stores with the continuing presence of comparable chains in the same sectors (Barnes & Noble, Kohl's, JC Penney, Best Buy, and Bed Bath & Beyond). Using geographically

[1] We will occasionally use the term "externalities" to refer to the impact of externalities, to avoid unnecessarily impenetrable language.

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detailed data on establishments we find robust, positive spillovers at the short distances relevant for municipalities - half of the towns in the country span less than 3 miles at their widest point.[2] After the disappearance of a big-box store, nearby businesses flounder. There are fewer of them, and they employ fewer workers. Based on data on household shopping behavior we establish that this happens in part because when consumers stop visiting the big-box store, they also visit nearby stores less frequently than they would have otherwise. This decline is absolute, and not just relative to stores close to big-box stores that did not close.

We then explore how local governments deal with these externalities. Externalities are a common justification for government intervention. If local governments are aware of the spillovers we observe, we might expect those local governments that face the largest positive externalities to be the ones that address them by implementing the most generous development policies. We test this hypothesis by exploiting the fact that some localities have shapes that allow them to capture more of these local spillovers, under the assumption that local governments care disproportionately about economic activity within their borders. We show that while the boundaries of the resident town do not affect the size of a big-box store's spillover at a given radius (i.e. the economic spillover), the "right" geographical shape of the locality makes it more likely that these positive spillovers are contained within city limits. Localities with more compact shapes are more likely to target retail development with subsidies and tax expenditures, conditional on total area. These results hold when we instrument for the shape of the political unit using a variety of legal and geographical impediments.

Of course, the right to locate near one of these large retailers is generally well-defined. The

[2]We use the terms "town," "municipality," "city," and "locality" interchangeably except where specificity is required to identify which data we use.

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Coase theorem would predict that, in such a situation, the private sector should provide mechanisms for internalizing these spillovers. In fact, prior research has shown that foot-traffic generating big-box stores are heavily cross-subsidized through private mechanisms like shopping malls, in which they often pay heavily reduced rental rates or even no rent at all (Gould and Pashinigan, 1998; Gould, Pashigan, and Prendergast, 2005). Brooks and Strange (2011) find that large retailers disproportionately benefit from quasi-private organizations like Business Improvement Districts (BIDs) as well. We show that cities that are more likely to offer public retail development subsidies are less likely to have these private and quasi-private incentive provision systems. Public internalization mechanisms crowd out the private-sector ones.

These estimates are important for several reasons. In a recent International City/County Management Association (ICMA) survey (ICMA, 2009), roughly 40% of local governments in the United States reported focusing on retail incentives to spur economic development. Be it through tax increment financing, income and property tax credits, land use subsidies, sales tax rebates, infrastructure assistance, or any of a myriad of other mechanisms, the goal is the same: to make it more attractive for large retailers to set up shop in town. The justification for local development efforts is that these retailers produce positive externalities for the existing local businesses and promote the arrival of new ones, increasing tax revenue and employment.

The amount of public spending involved in these types of subsidies is large and economically significant. Local governments in the St. Louis area, for example, provided over $5.8 billion in local subsidies over the past 20 years (East-West Gateway Council of Governments, 2011), of which some 80% went to retail stores. Since 1986, Chicago has "lost" $5.5 billion in revenue from committed tax increment financing districts (TIFs) (Orr, 2013). The perceived impact of these subsidies has made them one of the most common development strategies used by policymakers at

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the local level, leading to a developing regulatory push for increased transparency regarding their use and cost (Government Accounting Standards Board, 2014).

Despite the ubiquity of local economic development policies, earlier research expresses skepticism about the ability of governments to perceive and target true local externalities (Glaeser and Gottlieb, 2008). This paper sharpens this argument by demonstrating that even if governments are capable of identifying and targeting true externalities, government intervention may crowd out private mechanisms aimed at the same goal. In this sense, this paper provides rare evidence on the applicability of the Coase theorem: the situation we study features demonstrable external effects, observable variation in the level of public and private efforts dedicated to addressing them, but nonetheless similar economic outcomes.

This work also relates to the literature on the size of spillovers in the retail trade. Previous research has found fertile ground in the expansion of Wal-Mart, the largest firm by number of employees in the United States. Basker (2005) explores the impact Wal-Mart store openings had on local employment, and finds that while overall retail employment rises after a store opening, it does so by much less than the number of workers at the Wal-Mart store itself. Neumark, Zhang, and Ciccarella (2008) paint an even starker picture, of total retail employment dropping after Wal-Mart enters a market, suggesting particularly strong negative spillovers for other retail firms. This suggestion is supported by Jia (2008), who finds that almost half of the decrease in the number of small discount stores that occurred from the late 1980s to the late 1990s was driven by Wal-Mart's expansion, and by Merriman, Persky, David and Baiman (2012)'s case study of a Wal-Mart opening on Chicago's West Side. That is, not only do competing retailers shrink their work force, some are forced out of business entirely. Arcidiacono et al. (2016) and Ellickson and Grieco (2013) confirm the existence of negative spillovers for Wal-Mart's expansion into the grocery industry specifically,

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but show that those mostly affected chain outlets.

On the other hand, Bertrand and Kramarz (2002) and Sadun (2015) find that entry barriers meant to deter large retail stores harm job growth and smaller, independent retailers, respectively. Benmelech, Bergman, Milanez, and Mukharlyamov (2014) observe, in a similar vein, that retail stores bankruptcies produce negative spillovers for other retail establishments. Positive externalities from economic activity nearby are also found in the broader agglomeration literature (e.g. Arzhagi and Henderson, 2008; Greenstone, Hornbeck, and Moretti, 2010). We bring an explanation to the table that unifies these contrasting views: whereas the overall spillover effects we find are positive, we show that directly competing retailers are harmed by the presence of a big-box store. This conclusion is similar to that of Haltiwanger et al. (2010), who study entry into the Washington, D.C., metro area, and may explain Schuetz's (2015) finding that big-box stores tend to locate near non-competitor big-box stores.

The overall positive spillovers we find do, of course, have very different implications for firms and policymakers, and we dedicate the second half of our paper to those implications. We establish that policymakers respond to the spillovers we observe by targeting retail development more aggressively in places where they are larger, and that this is at least partially superfluous because it crowds out private-sector action that accomplishes similar goals.

The paper proceeds as follows. In section 2, we demonstrate the ex ante similarity of our comparison neighborhoods and document the absence of pre-trends in the data. We then use difference-in-differences regressions to estimate the externality associated with a major big-box closing on outcomes for neighborhoods of differing sizes around the store. Relying on locally exogenous variation produced by national-chain bankruptcies, we find strongly negative overall effects that decay with distance. As independent confirmation of our results, and to establish the mechanism

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through which the observed spillovers arise, we therefore study consumer behavior. We find that after a big-box store closes, consumers rapidly and significantly reduce their visits to stores they used to visit on the same day as the big-box store. We also demonstrate here that consumers do not just substitute away toward stores near competitor big-box stores.

In section 3, we show that the size of these effects for political units (cities and towns) depends upon a city's geographic shape, or "compactness." While compactness does not affect the size of the externality as measured over fixed distances both inside and outside the political unit, it does affect the size of the spillover within a city's borders and, as a result, a city's propensity to offer retail development incentives. Section 4 shows that this propensity for public subsidies is negatively correlated with the presence of shopping centers and BIDs, suggesting significant crowd-out effects. Section 5 concludes with a discussion of our results, and why they matter for both the development of urban centers and for our understanding of the history of the city.

II Externalities from Big-Box Retail Stores

Between early 2007 and early 2008, CompUSA, a consumer electronics chain, ended all of its retail operations (McWilliams, 2007). Also in 2008, Mervyn's, a chain of department stores, filed for Chapter 11 bankruptcy and liquidated its stores in seven states (Dodes and McCracken, 2008; Mervyn's Brands, LLC, 2008). Later that year, Circuit City Stores entered bankruptcy and by January 2009 started closing its remaining Circuit City Superstores (Chang and Zimmerman, 2009). Around the same time, Linens 'n Things, a major chain of housewares stores, closed its remaining locations (Dealbook, 2008). Finally, in 2011, Borders Group, an international chain of book and music stores, applied for Chapter 11 bankruptcy protection (later converted to Chapter 7), could not find a suitable buyer, and liquidated its stores across the U.S. (Border's Group, Inc.,

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2011; Spector and Trachtenberg, 2011). Not all big-box retail chains in these industries suffered the same fate. The book stores of Barnes & Noble, the department stores of JCPenney and Kohl's, Best Buy's electronics stores, and Bed Bath & Beyond's housewares stores continued to operate. The bankrupt and surviving chains we study are similar in geographic dispersion, and we will see that despite suggestions in the popular press (e.g. Hamilton, 2008; Lowrey, 2011), features of the local business environment cannot account for their different fates. Instead, high-level strategic choices related to capital structure or online competition made by the firms may explain their survival or non-survival (e.g. Galuszka, 2008; West, 2009; Lowrey, 2011; Farrell, 2012). We use as our measure of bankruptcy timing the year in which the national chain filed for bankruptcy, so that our measure of timing is not driven by local idiosyncracies either.

In this section, we compare the economic activity around the surviving big-box stores to that around the locations of the now defunct stores, to produce estimates of the spillovers produced by the presence of these large retail stores. We do so by using data on establishments and the workers they employ, as well as data on consumer purchases. In this way we derive the estimates of interest from two independent data sources, from the supply and demand side of the economy.

II.a Nearby Establishments

We locate big-box stores and neighboring establishments by using commercial Esri Business Analyst data supplied by Harvard's Center for Geographic Analysis.[3] We calculate distances be-

[3]These geo-coded establishment data originate with InfoUSA, a business partner of Esri, which collects lists of establishments from phone directories, business filings, utility connections, press releases, web directories, annual reports, and other sources. InfoUSA then surveys each establishment by phone (between 12 and 18 million establishments per year) and collects, among other things, data on employment and industry.The data are annual and run from 2003 to 2012; for

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tween each establishment and the big-box retailers in our comparison sample, and we then aggregate outcomes in terms of store counts and employment for neighborhoods of varying radii around big-box retailers and for the city containing the big-box retailers. Although the Esri data are not produced by the federal statistical system, we believe that it is the correct choice for a variety of reasons. First and foremost, Esri includes address data, which allows us to work at a high-precision geographical level. It includes establishments not counted in the Census' County Business Patterns dataset (unincorporated or no employees) - someone teaching violin lessons might show up, for example. This broader measure of economic activity is appropriate for our purposes (and for many other purposes as well, which is why there is a market for these datasets), although there may be systematic scale differences. We always include fixed effects, and our estimates are in percentage terms. In addition, our results stand even if one worries about the quality of the data. We use establishment counts as a dependent variable, and as long as the noise in the Esri data is not correlated with a chain eventually becoming defunct, a type of correlation that strikes us as wildly implausible, the noise merely inflates our standard errors. That said, we do not believe that even measurement error is particularly severe.[4]

Now, although the above mentioned combinations of chains are in similar industries, the establishments of comparison pairs (e.g. Borders and Barnes & Noble) were not necessarily identical

selected summary statistics, see Appendix Table 1.

[4]The correlation between Census Business Patterns data on the number of establishments in all zip codes, in zip codes with over 50 establishments, and in the zip codes that one of our big-box stores calls home and the comparable Esri numbers is over 0.9 for each year in our dataset. We confirm our basic results based on Census data in Appendix Tables 2 and 3. Part C of the Data Appendix discusses the Esri data more extensively.

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prior to chain-level bankruptcies. Table 1 addresses this concern. The table shows estimates of the coefficient on a dummy variable indicating whether a store would eventually go bankrupt in the following regression equation:

$$\Delta\ ln(Stores,\ Employment)_i = \alpha + (Bankrupt)_i\beta + \gamma_t + \varepsilon_i,$$

where γ_t are fixed effects for each big-box store type (clothing, for example), and β measures the difference in the growth rate of the number of stores or total employment in the concentric areas around big-box retailers that disappeared and that survived. Observations are neighborhoods around big-box stores. The first two columns show pooled estimates of the log increase from 2006 to 2007 in the number of stores and level of employment for radii of increasing size (0.5, 1, 2, and 3 miles) around defunct stores compared to non-defunct stores. There are no significant differences between trends in the number of area stores or area employment between defunct and non-defunct stores. As Columns 3 and 4 show, the same holds true for differences in the growth of the number of stores and of employment within the different radii over the preceding 5-year period from 2003 to 2007.

Perhaps even more convincingly, Figure 1 graphically represents the evolution over time of the log number of establishments within a mile of defunct big-box stores relative to the number of stores within one mile of non-defunct chains, with year-store type and neighborhood fixed effects removed. Neighborhoods in our sample that were home to ultimately defunct big-box stores followed similar trends to the control neighborhoods until the national-chain bankruptcy, when they started diverging.[5]

[5]Similarly, Appendix Table 4 provides a basic pre-bankruptcy comparison of the level of economic activity in 2003 and 2006 around defunct and non-defunct big-box stores that shows no

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Of course, pre-trends may mask heterogeneity across chain locations in susceptibility to subsequent shocks. To address this possibility, we exploit the geographic detail of our data to estimate the impact of closures within zip codes. Even when we include *zip-code-by-year fixed effects* (i.e. dummies for 02138 in 2010, 20037 in 2010, 02138 in 2011, etc.), and thus control for arbitrary zip code level trends, we recover a similar result.

Having confirmed the similarity of our treatment and control neighborhoods, we can now turn to our baseline estimates of the spillovers of interest. Table 2 provides these. The equation we estimate here is the following:

$$ln(Stores,\ Employment)_{i,t} = \alpha_i + \gamma_t + (Post\ x\ Bankrupt)_{i,t}\beta + \varepsilon_{i,t},$$

where α_i are fixed effects for each big-box location neighborhood, γ_t are year-store type (or year-zip code area) fixed effects, and β measures the differential impact of a post-closure year on neighborhoods where the big-box retailer chain has closed. The identifying assumption is that neighborhoods around defunct and non-defunct retailers would have fared similarly, were it not for the nationally determined closures. Of course, if national-level big-box bankruptcies are correlated with sharp breaks in (very) local trends within zip codes, that would still introduce bias.

Columns 1 and 2 in Table 2 show that after a chain-wide bankruptcy, the number of stores within a quarter-mile radius of a defunct store decreases by some 8 to 11%, while employment drops by 7.5 to 13% compared to the same area around non-defunct stores.[6] To control for any

significant differences. Appendix Table 5 shows that this also holds true for a range of other neighborhood characteristics.

[6]Appendix Figure 1 visualizes our empirical strategy for two electronics stores in Massachusetts.

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potential regional differences, we repeat this exercise controlling for arbitrary trends at the zip code level by including zip-code year fixed effects. The impact remains large and significant with these controls, although the effect size is smaller.[7] This fact is consistent with a greater weight on overlapping neighborhoods when using solely within zip code-year variation. To test for the importance of overlapping neighborhoods in our general results, in row 3 we re-estimate the effect using only zip codes that contain at most one store per store type. These estimates lie quite close to the baseline effects.[8] These results are not driven by a shift toward the neighborhood surrounding the big-box store's direct competitor. This is most easily seen in the next subsection, in which we study consumer shopping behavior directly.

In the median neighborhood these effects involve around 20 stores and 300 employees: fairly large numbers, of a magnitude reminiscent of the results found by Jia (2008) and Neumark et al. (2008).[9] They do not, of course, materialize instantly. Figure 1 shows, for the 1-mile radius, that the effect builds gradually over a period of multiple years. The estimates are precise, and we can

[7]Appendix Figure 2 visualizes our empirical strategy for two stores in the same zip code.

[8]We use Coarsened Exact Matching (Iacus et al., 2012) in Appendix Table 6; the results are both qualitatively and quantitatively similar. The same holds when we use fixed, never overlapping geographies, as in Appendix Table 7.

[9]An alternative scaling of the coefficients lets the effect of a big-box closing vary with a store's estimated sales volume. To estimate this effect, we run a regression of $ln(Stores) = \alpha_i + \gamma_t + \textit{(Post x Sales Volume)}_{i,t}\beta_1' + \textit{(Post x Bankrupt)}_{i,t}\beta_2' + \textit{(Post x Bankrupt x Sales Volume)}_{i,t}\beta_3' + \varepsilon_{i,t}$ at the 1 mile radius. We estimate β_3 at a statistically significant -0.011, with a standard error of 0.004. The typical big-box store in our sample had sales of roughly \$7.5 million in 2007, with an interquartile range of \$4.0 million to \$19.4 million. This implies that a big-box store closing at the 75th percentile led to a 1.7 log point greater decline in neighborhood stores than one at the 25th percentile.

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reject the null of no effect at the 1% level.

Columns 3 through 8 show estimates for the number of stores and employment within a half-mile, one-mile and two-mile radius. The most striking feature of this table is significant decay in the effect size of the spillovers we observe as we move farther away, reminiscent of Campbell, Giglio and Pathak (2011). Within one mile, the number of stores decreases by 3% post bankruptcy while within two miles, we see a drop in the number of stores that is just below 2%.[10][11] Employment levels show a similar declining pattern.[12] It is not immediately clear from these numbers whether the effects are driven primarily by shifting economic activity across locations or by changes in the aggregate number of establishments and employees, but the effects we observe at the very local level are strongly negative and are not balanced out by positive effects somewhat farther out. [13]

[10]We also estimate the one mile radius effect on the number of stores using data for one Combined Statistical Area at a time. Roughly 70% of these CSA-level estimates are negative. The mean estimate is -0.027, with a standard error of 0.006, and is highly statistically significant. The median estimate is -0.021. These effects are quite close to our baseline.

[11]It is possible to control for potential differential trends along demographic lines. We use data from the 2000 Census at the zip code level to create median-house-price-by-year dummies, percent-black-by-year dummies, and unemployment-rate-in-2000-by-year dummies. Without these controls, our baseline result in Table 2 at 1 mile was 0.030 ($SE = 0.005$); with these controls it is virtually identical at 0.029 ($SE = 0.005$).

[12]Another possible set of specifications uses concentric rings rather than concentric circles. These specifications return very similar results.

[13]In Appendix Table 8 we disaggregate our estimates by type of big-box store. The externalities we observe as created by the different store types are comparable both qualitatively and quantitatively, and we focus our attention on pooled estimates in the rest of the paper.

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Just like the impact felt by nearby stores differs from that felt by stores farther away, different types of stores are affected differentially as well. Table 3 sheds some light on the differences between spillovers affecting four mutually exclusive pairs of store types: businesses that depend on foot traffic versus businesses that do not; firms that do not compete with the big-box stores we track versus firms that do; new entrants versus incumbents; and businesses in areas with high versus relative low levels of Internet access. It shows results for regressions similar to those in Column 3 of Table 2 (effects within a half-mile radius), with year-store type fixed effects. One would expect that businesses that depend on foot traffic are affected more heavily by the disappearance of a nearby anchor store. That is correct, as we see in Columns 1 and 2: 9% of businesses that rely on foot traffic disappear, while firms that do not are not significantly affected.[14] We see an even starker contrast in Columns 3 and 4. Firms that used to compete with one of our big-box stores flourish after said store's disappearance, growing in number by 4%. Firms that did not are hit with the negative externality we observe in the aggregate, and their number decreases by 3%. In the case of the big-box stores we focus on, most stores do not compete with our big-box stores, of course, which explains the positive aggregate effects we measure. It is also, we believe, a potential explanation for the discrepancy highlighted in the introduction between the negative overall spillovers produced by the opening of a Wal-Mart, which competes with many existing

[14]It is also true, as Column 1 and 2 of Appendix Table 9 show, that bankruptcies of stores in more "walkable" locations or with better access to public transportation produce greater spillovers. Interestingly, how close a store is to the city center (Column 3) does not appear to affect the size of externality. The preference that both anchor stores and other retail stores have for downtown locations (Billings and Johnson, 2012; Zhou and Clapp, 2015) illustrates that they do not appear to fully internalize the spillovers they produce.

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businesses, and the positive spillovers observed in the literature around other large retailers, which compete with a smaller share of incumbent businesses.

A different dichotomy of interest is that between incumbent firms and new entrants. In Columns 5 and 6 of Table 3 we see that both categories suffer when a neighborhood suddenly loses its big-box stores. The number of incumbent firms decreases by some 2.5%, while the number of entrants goes down by even more: a little over 3%. Given that new entrants account for only 17.3% of establishments, in absolute terms the lion's share of the burden of big-box store disappearances is borne by existing firms. Columns 7 and 8 confirm the importance of online competition in recent years: the spillovers we observe are larger in states with higher levels of Internet access, possibly because more stores there were already on the brink of bankruptcy.

II.b Consumer Behavior

Our spillover measures so far have been based on data on establishments. As a check on their validity, and to assess the mechanism through which these spillovers arise, we turn our attention to the customers of these establishments. To do so we use household-level data on shopping behavior from Nielsen. The data describe the purchases of retail goods of some 100,000 households in 20,000 zip code areas between 2004 and 2009, for a total of 300 million transactions. While the stores visited by these households are identifiable in the sense that visits to the same store can be tracked, Nielsen shields the precise names and locations of the stores, forcing us to rely upon a characterization of the store's line of business and the first three digits of its zip code. The stores we look at here are the electronics stores, department stores, and housewares stores, as Borders went bankrupt too late for us to be able to study the impact of its bankruptcy using these data.

Our empirical strategy is as follows. We focus on households that live in a three-digit zip code

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in which one of the ultimately defunct big-box stores is located. We can only identify large categories of stores, so within this group of households we treat households who visited an electronics, department or housewares store as households who visited a store of the corresponding ultimately bankrupt big-box chain of interest on that day. This leaves us with between 500 and 1,000 households for each of the chains. The variable we track is the ratio between the monthly number of stores visited on the same day as the imputed ultimately defunct stores, and the number of stores visited on other days. If this ratio goes down, it means that stores that are visited on the same day as stores we have identified with CompUSA, Mervyn's, Circuit City, and Linens 'n Things are being frequented less often or even close down entirely. This ratio went down after the chains we study disappeared (Appendix Table 10). Appendix Table 11 shows the results of simple regressions of this ratio on a post-bankruptcy dummy, with household fixed effects included; they show consistently significant decreases in this ratio. These estimates, however, underestimate the full impact of bankruptcy on surrounding retail. We can observe how this happens in Figure 2. After bankruptcy, households take a while to adopt new shopping habits, but the stores they used to visit whenever they visited the now-bankrupt big-box store clearly suffer the consequences of their big neighbor's demise.[15] In some cases they see a decrease in the number of visits of as much as 20% relative to non-affected stores. Household fixed effects do not change our results qualitatively.

This approach also helps us distinguish between different possible interpretations of the results presented in Table 2. There we compared outcomes for neighborhoods around defunct chains to

[15]The exact bankruptcy dates we use are May 2008 for Linens 'n Things, July 2008 for Circuit City and CompUSA, and October 2008 for Mervyn's. There is some variation in the closing dates of individual stores, but our choice of bankruptcy date only affects the locations of the vertical lines in this figure, not our estimates of monthly shopping behavior as such.

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neighborhoods around competitor chains. This difference-in-difference strategy could not distinguish between declining store counts around defunct chains and rising store count around competitor chains. Here, on the other hand, we focus on households that live within a zip code in which one of our ultimately defunct chains was located. In other words, any store that an individual near a Circuit City visited on a day he made an electronics purchase was tagged as a same-day store. The decline in same-day store purchases by this individual, relative to non-same-day store purchases, cannot then reflect an increase in visits to a nearby Best Buy. It must reflect a decline in visits to stores near all local electronics stores - or a drop in traffic to the defunct neighborhood. Note that we are not suggesting that this excludes the possibility of substitution effects explaining part of our establishment-based estimates; it simply confirms that at least some of the decline in economic activity around defunct store is absolute.

This result is confirmed by another test, in which we estimate the same difference-in-difference regressions in Table 2 on only the set of neighborhoods around ultimately defunct stores. These regressions identify the impact of the bankruptcy off differential timing: Borders went bankrupt later than the other chains. These regressions produce somewhat smaller, but highly significant declines of 1.5% (standard error of 0.4%) due to bankruptcy.

Now that we have demonstrated the existence of big-box store externalities, and what drives them, we ask: how are they related to town characteristics? Are local governments aware of the sizable externalities we have observed, do they care about them, and do they use policy to attract the stores that create them to their towns?

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III Externalities, Local Geography, and Local Government

The sizable spillovers from big-box retail stores that we have identified have not gone unnoticed by policymakers. In this section, we analyze the development incentives offered by local governments to address them. The shape and boundaries of localities affect the extent to which externalities are internalized (Hayashi, Nishikawa, and Weese, 2016), and we will see in this section that that certainly holds true here. As a consequence, local-government development policies are related to town shape as well, confirming the finding of Felix and Hines (2013) that geography plays a central role in shaping such policies.

It is not just academic research that supports the idea that the geographical shape of local government units is an important driver of development policies. Many city governments - 73% of them according to ICMA (2009) - claim to perform a formal cost-benefit analysis prior to offering business incentives. There is a robust market for economic-development consultants, and the models used by these consultants alert city governments to the importance of city boundaries: these so-called economic impact models often emphasize the importance of taking into account how the shape of a town or other political unit interacts with spillovers to determine the incidence of the externalities created by government subsidies. The Federal Reserve Board of Governors' Fiscal Impact Tool (2003), for example, explains that "[p]rojects located on the edge of a town, closer to the shopping and transportation access of a neighboring community, will likely experience greater-than-average retail leakage to that other community." The Bureau of Economic Analysis' (2012) Regional Input-Output Modeling System, which produces regional "multipliers" that are meant to be used to estimate the economic impact of projects within a region, explicitly models "money leaking out of the economy." Other economic impact models also take into account factors like

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the share of sales tax receipts in the city as a share of total receipts within a driving radius (e.g. Kennington, 2011, and more generally all of the tools discussed by Morgan, 2010), reflecting the importance of town shape in capturing externalities.[16]

III.a Town Shape and Externalities

To speak to these policy-relevant concerns we move to town-level regressions, where we use Census Incorporated Places to define towns. This allows us to focus on the externalities of direct interest to local policymakers, instead of the radius-based ones highlighted before. Table 4 shows these results in Columns 1 through 4, while Columns 5 through 8 show results for the 1-mile radius in an otherwise identical specification. In Columns 1 and 3 we see that the negative impact on the number of stores and total employment in a city after one of our big-box stores closes is sizable and significant, as it was within the 1-mile radius around the store. On average, a big-box bankruptcy in a city leads to a 0.7% decrease in both the total number of stores and employment for that city, or the disappearance of 28 establishments and 333 jobs.

Big-box store externalities thus extend to city-wide effects. This is intuitive, given that according to the Harvard Center for Geographic Analysis 75% of cities in our sample span less than 4 miles at their widest and longest points. A consequence of the comparable scale of spillovers and municipalities is that the precise shape of municipalities matters. We therefore turn our attention now to estimating how much larger the town-level spillovers from big-box stores are in towns shaped in ways that allow them to capture most of the total radial spillover as compared to towns shaped differently.

The measures of town shape we will use are all based on the same basic notion of "compact-

[16] An example of this phenomenon in our data is visualized in Appendix Figure 3.

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ness." Imagine a city shaped like a perfect circle. If a big-box store is placed in the middle, that city will capture nearly all of the benefit. On the other hand, if we place a big-box store in the center of a city shaped like the letter "L", the benefit is shared with neighboring localities. With this intuition in mind, we employ five measures of compactness, most of which originated in the political-science literature on the shape of congressional districts: the ratio of town area to the product of the maximum width and length of the town, which we will refer to as the rectangular area fraction (RAF); the ratio of town area to the area of the smallest circle that can enclose the district, as proposed by Reock (1961); the ratio of town area to the area of a circle with the same perimeter, first suggested by Polsby and Popper (1991); the ratio of the town's perimeter to that of a circle with the same area as the town, from Schwartzberg (1966); and the ratio of town area to the area of its convex hull, the convex ratio. These measures are strongly correlated, and for clarity's sake we aggregate them into an index: the different measures' first principal component.[17]

In the remainder of Table 4, we relate the compactness index to the the size of radius and town-level spillovers by estimating the following equation:

[17]The first principal component has an eigenvalue of 3.9, and accounts for 79% of the variance. The five compactness concepts are illustrated in Appendix Figure 4. Appendix Tables 3 and 12 show the main results separately for each compactness measure. Appendix Figure 5 shows the RAF for the Boston metropolitan area as an example. Appendix Figure 6 demonstrates that the average distance from stores to city limits is greater in more compact cities, as one would expect. As one moves from the low end to the high end of the compactness range, the expected average distance to city limits goes up by a little short of half a mile. That is precisely the order of magnitude of the distances within which most of the externalities take place.

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$$ln(Stores,\ Employment)_{i,t} = \alpha_i + \alpha_t + (Defunct)_{i,t}\beta_1 +$$
$$(Defunct\ x\ Compactness\ Measure)_{i,t}\beta_2 + \varepsilon_{i,t},$$

where α_i are town (or neighborhood) fixed effects, α_t are store-type-year fixed effects, Defunct replaces Post x Bankrupt for ease of notation, and β_1 and β_2 measure the size of the spillover and the interaction of the spillover and the compactness ratio. Column 2 shows that the number of stores that go out of business after a big-box bankruptcy is significantly higher in compact towns than in non-compact ones, even though the radial externality does not differ significantly, as shown in column 6. Columns 4 and 8 shows very similar results for the level of employment. What this means is that even though the radial externalities are the same for big-box stores in compact and non-compact towns, compact towns derive larger town-level externalities from their presence.[18] The absence of a radial effect suggests that the interaction effect for the town is causally driven by a town's shape. To give a sense of the magnitude of the effect, let us look at a concrete example. As one moves from the less compact city of Silver Spring, MD, to more compact Arlington, VA[19], our estimates in Columns 2 and 4 suggest that the number of within-town stores that disappear after a bankruptcy increases by 1.7 percentage points, while the employment drop grows by 1.2 percentage points.

[18]Appendix Table 13 makes this point in a different, perhaps more explicit way: it shows that town-level externalities are greater near town borders for more compact towns.

[19]Arlington is a county, but its county government is its only general-purpose local government.

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III.b Town Shape and Development Efforts

Having established that less compact localities facd smaller externalities within their borders, let us see if they also make less of an effort to attract big-box stores. This would make intuitive sense, whether local government officials want to get reelected, maximize tax revenue, or create jobs for local residents. Retail subsidies are a particularly popular way to attract business activity, as 40% of local governments report focusing on them (ICMA, 2009). The question we ask here is whether compact towns are more likely to do so. We address this question by estimating the following equation:

$$\textit{Retail Focus Development}_i = \beta_0 + \textit{Compactness Index}_i\beta_1 + X_i\gamma + \varepsilon_{i,t},$$

where the dependent variable is an indicator for retail focused development in the ICMA 1999, 2004, and 2009 Economic Development Surveys, the compactness index is as described above, and X_i are various control variables for the municipalities. Column 1 of Table 5 shows the results of these regressions without controls, and as expected, more compact cities are strongly associated with increased development focus under most measures. Column 2 reports the coefficients controlling for population decile dummies, state dummies, the logarithm of the median house price, and the percentage of the population that does not speak English, is in poverty, and holds a bachelor's degree. Column 3 controls for the logarithm of the total sum spent on development strategies as well as the total number of development strategies (retail and non-retail) pursued by the town in question. Column 4 controls for whether towns have a large retail tax base and a sales tax. Column 5 reports the coefficients controlling for Combined Statistical Area (CSA) fixed effects. Across all of these controls and measures, there is a consistent relationship indicating that more compact cities - cities more likely to capture retailer externalities - are most likely to offer programs aimed

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at promoting them. Note that Column 5 indicates that this holds even for towns *within* CSAs. This is an important result, because it means that local governments are sensitive to the size of the externalities they face when setting development policy; the extensive battery of town features we control for makes it hard for us to imagine that this relationship is a purely coincidental one or one driven by omitted-variable bias. Let us once again compare Silver Spring, MD, to Arlington, VA, to clarify what this means in more concrete terms. As one moves across the Potomac to Arlington, VA, our estimates suggest that the likelihood of the town focusing on retail subsidies increases by between 4.5 and 12.1 percentage points.

It is, of course, possible that this spillover-to-subsidy relationship is spurious, despite the anecdotal evidence we have presented on economic development consultancy services. For example, compactness could be correlated with a confounding factors, such as local corruption, that both make retail subsidies more likely and correlate with the size of retail spillovers. We address the possibility that this confounding channel accounts for the entire relationship in three ways.

First, a number of instrumental-variable estimates of the relationship between compactness and retail focus confirms our OLS results.[20]

Second, we check whether compact and non-compact towns act similarly when it comes to non-foot traffic industries. Our causal hypothesis predicts differences only in retail, whereas one might suppose that an omitted factor would manifest itself in other subsidies as well. Panel A of Appendix Table 14 shows regressions similar to Table 4 for retail and non-retail development programs in the 2009 ICMA wave. We find no relationship between compactness and subsidies for agriculture, nonprofit institutions, tourism, transportation, technology, and "other" activities.

[20]These estimates are discussed in detail in the Instrumental-Variable Estimation Appendix.

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Again, this suggests a causal relationship between foot traffic spillovers and retail development. To strengthen this second point we turn to a database of actual subsidies. This database, previously used in Story, Fehr, and Watkins (2012) and Ossa (2015), was collected by Good Jobs First (GJF), a non-profit organization in Washington, D.C., in its Subsidy Tracker, and reflects subsidies provided to corporations across the United States. We include towns in which GJF observed at least one subsidy program since the start of its dataset in 1976, and construct dummy variables for each of these local governments that indicate whether there was at least one retail and at least one non-retail program in the locality. Although these data are subject to selection bias (though not obviously selected on compactness), they have the advantage of being derived from behavior, not expressed intent. In Panel B of Appendix Table 14, we show the results of a regression of these dummy variables on our compactness measure. Our results confirm our central findings in this section: more compact towns, those that can capture a larger share of the externalities big-box stores create, see more of a focus on retail development than less compact ones.

Finally, we note that compactness should be a larger determinant of subsidies in small jurisdictions. While we continue to control for area as in Table 5 and Appendix Table 14, our regressions in Appendix Table 15 also split the sample at the median jurisdiction size. We then report the impact of compactness, again using OLS and IV, for large and small cities. As predicted, we find a much larger effect in towns where the border effects are likely to be larger. It also implies that all else equal, more compact towns should be more likely to provide subsidies close to town borders, where the potential for "leakage" is particularly high. Appendix Figure 7 shows that this is indeed the case.[21]

[21]Further suggestive evidence comes from the fact when we include the number of nearby towns as a control variable, a rough test for the impact of competition between jurisdictions, the coeffi-

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The combined weight of these results, we argue, indicates that cities with larger potential spillovers are more likely to offer retail subsidies. An alternative explanation of the facts requires something correlated independently with a number of instruments and compactness itself that is not also correlated with city-level income, house prices, population density, education levels, and municipal expenditures and revenues. This omitted variable would also have to affect retail subsidies alone, not other industries, and operate primarily in smaller towns where more of the jurisdiction is proximate to the border. Given this evidence, and the anecdotal evidence from our survey of economic development consultants, the most compelling interpretation is that compactness and spillovers indeed causally affect local government behavior.

IV Private Subsidies and Compactness

The previous section demonstrated that measures of city compactness are strongly correlated with the size of spillovers falling within city borders. Local governments are seemingly aware of this, and more compact municipal governments target retail development more aggressively. This takes us back to Coase's observation: that well-defined property rights and limited transaction costs should allow for private-sector mechanisms to internalize externalities. As long as the land surrounding potential big-box stores can be controlled by the same party, or the owners of nearby stores can resolve their collective-action problem, there ought to be no need for government intervention. Such private-sector mechanisms have indeed come to fruition. In this section we study two of them: shopping malls and Business Improvement Districts (BIDs). The central questions

cient on that variable is positive and significant. It would be of interest to explore what effect frictions impeding land assembly, which effectively reduce the number of developable sites (Brooks and Lutz, 2016) while increasing their price, have on the likelihood of retail subsidy programs.

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we address here are whether these organizations actually subsidize big-box stores, and whether public subsidies crowd out these private alternatives.

IV.a Shopping Malls

Gould et al. (2005) demonstrate that anchor stores in shopping centers pay substantially discounted rents, with 73% of anchor stores in their sample paying no rent at all to the mall owners. [22] This steep discount is recouped by shopping center ownership via the premium charged on rental space to non-anchor stores that hope to benefit from the foot traffic generated by the anchor stores (Benjamin et al., 1992; Konishi and Sandfort, 2003). As an example, food court tenants in our sample pay over $60 per square foot. In this manner, joint ownership of both the anchor and non-anchor stores in the form of a shopping center or mall provides a private-sector mechanism for internalizing the externality effects. This internalization is observed directly in Liu and Liu's (2013) study of REITs, which are strongly affected by bankruptcies involving their anchor tenants. To test whether this private mechanism is affected by local-government subsidization, we analyze data on the prevalence of shopping centers by location. The data, provided by Esri, contain information on over 7,000 geo-coded malls and shopping centers. We have been able to match more than 6,000 of these to incorporated places for which we have data.[23] We test the relationship between shopping center prevalence and compactness by estimating the following equation:

$$\textit{Shopping Mall Indicator}_i = \beta_0 + \textit{Compactness Index}_i\beta_1 + X_i\gamma + \varepsilon_{i,t}$$

[22] We present similar results in Appendix Table 16, based on data from ICSC (2004). Anchor stores pay much lower rental rates than other stores no matter how large they are, for an average of $1.84 per square foot, as compared to an operating cost of $16.37 per square foot for the mall.

[23] Selected summary statistics are shown in Appendix Table 17.

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The results for this estimation are presented in Columns 1 through 4 of Table 6. The dependent variable is a dummy for the existence of a shopping center, while the explanatory variable is our compactness index, except for in Column 1, where it is a dummy for retail focus in the ICMA surveys. Each regression controls for land area.[24] Column 1 shows us that towns with more of a retail focus have fewer shopping malls: the probability of having at least one is reduced by seven percentage points there. Such towns are more compact than others, and it should not come as a surprise that Columns 2 suggests that more compact towns are indeed significantly less likely to contain a shopping center. This result suggests that the aggressive attempts to attract retail carried out by compact towns make it less likely for those towns to harbor shopping centers.

Columns 3 and 4 in Table 6 explore where these marginal shopping malls are, and finds that they are largely located within 2 miles of the city limits, as expected: those are, after all, the areas that the governments of compact towns will disproportionately focus their subsidies on compared to those of non-compact towns.[25] Columns 5 and 6 study the intensive margin: the log number of malls/centers, and the log number of stores in malls/centers, conditioning on the existence of

[24]Appendix Table 18 shows that the results are robust to the inclusion of additional control variables, while Column 1 through 4 of Appendix Table 19 show similar results for our different compactness measures instead of the aggregate index; see the Instrumental-Variable Estimation Appendix for IV estimates.

[25]The effect of compactness on the likelihood of having a mall decreases with distance to the border examined. A 1 standard deviation increase in compactness makes a town 4.1 percentage points less likely to have a town within half a mile of the town border. The same increase in compactness is associated with a 1.4 percentage point decrease in the likelihood of having a mall 0.5-1.0 miles from the border, a 0.5 percentage point decrease in the likelihood of having a mall $1.0 - 1.5$ miles from the border, and a 0.2 percent decline in the likelihood of having a mall $1.5 - 2.0$ miles from the border. The effect becomes statistically insignificant at greater distances.

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a major shopping center. They demonstrate that, even conditional on having a major shopping center, more compact towns will have fewer such centers and fewer stores within such centers. Note that these results hold even for different towns within the same CSA. In addition, Appendix Figure 8 shows that in more compact towns, the percentage of stores that find a home in shopping centers is also lower. All in all, the evidence for public-sector development initiatives crowding out shopping centers is strong and consistent.[26]

IV.b Business Improvement Districts

The second private-sector mechanism that can substitute for local-government provision of incentives to internalize externalities is the establishment and operation of Business Improvement Districts (Brooks, 2007). In some jurisdictions, property owners in commercial neighborhoods can agree to found such districts by majority vote. BIDs then have the power to tax and spend. Assessments are mandatory and typically used for the provision of public goods, such as safety (Brooks, 2008). That said, the benefits from BIDs do not flow to all participants equally. Brooks and Strange (2011) find that it is typically the large anchor stores participating that reap the direct monetary benefits, much like in a shopping mall.

We would therefore expect local-government subsidies to big-box stores to affect BIDs in a way similar to what we saw for shopping malls, and the tests we run are similar as well. Instead of our shopping mall-related dependent variables, we create a dummy variable that indicates whether a town contains a BID using data from the Rose Institute of State & Local Government (2013). We

[26]Shopping centers are certainly not perfect substitutes for decentralized street-oriented shops from the perspective of either the local government or the consumer, for a variety of reasons including esthetics and market structure, and we would not expect full crowd-out for that reason alone.

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then estimate how much less likely a more compact town is to see a BID arise, and as before we present our results both with and without controls. What we find is, as Column 7 of Table 6 shows, that BIDs are more likely in less compact towns. To return to our example from the previous section: as one moves from a city a compact as Silver Spring to one as compact as Arlington, the likelihood of encountering a BID increases by 14 to 20 percentage points,[27] confirming our findings regarding shopping malls.[28] It is striking that these two private-sector alternatives to direct government subsidies both appear to be crowded out by said subsidies, in spite of the quite different ways in which they are brought to life.

V Discussion and Conclusion

We have reported three central empirical findings. First, using detailed data on households and establishments, we established that there are large, positive local externalities associated with big-box stores that increase the number of non-competing businesses nearby, as well as employment at those businesses. Second, local governments use subsidies and tax expenditures to attract big-box stores to their towns more aggressively the more the externalities are contained within their locality. And third, we observe that these public programs make private-sector initiatives through which the externalities can be internalized, such as shopping malls or BIDs, less common. The latter finding is consistent with some standard crowd-out models, although it is certainly not an implication of all of them.

[27]Column 5 and 6 of Appendix Table 19 show similar results for our different compactness measures instead of the aggregate index.

[28]It should therefore not come as a surprise that there is, indeed, a BID in Arlington (in Ballston, specifically: `http://www.ballstonbid.com`), whereas Silver Spring, to the best of our knowledge, does not have one.

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The relative efficiency of local government subsidies versus private sector mechanisms is a difficult question. In Table 4, we saw that compactness was not significantly linked to the economic externality at fixed distances. This suggests that public subsidies do not impact the efficiency of the retailer's location choices as measured by these external effects. It is important to realize that this result - a manifestation of Coase's (1960) insight - is only true in the aggregate. The localities where big-box stores end up may well benefit from well-crafted development policies if crowd-out is not complete and the subsidies offered are smaller than the spillovers realized.[29]

A rough back-of-the-envelope calculation shows that this may frequently be the case. Using Esri's assessments of sales, and following an approach similar to that used in Section II, we estimate a 3.1% increase in sales within a mile of one of our big-box stores, or a $35 to $40 million annual swing for the median neighborhood. The externality is even bigger for the median town as a whole: north of $100 million a year. Now, this is total sales, not just increased income to local store owners, tax revenue, consumer surplus, and salary hikes, but it dwarfs the average subsidy paid to attract a Wal-Mart Superstore,$2.8 million (Mattera and Purinton, 2004). It also dwarfs the cross-subsidy received by anchor stores in shopping malls, which, at $15 per square foot equals around $2 to $3 million for the typical anchor store.

But these numbers do not come close to painting a full picture of the role these public policies play in shaping the living and shopping environments of the hundreds of millions of people who live in urban and suburban areas in the United States alone. Although entrepreneurs grew aware

[29]Even with complete crowd-out there is one party that benefits: the non-big-box stores that now see taxpayer money being used to subsidize the big-box stores instead of their own inflated rent payments. This incidence result is particularly interesting given the political rhetoric that often surrounds these subsidies (e.g. Mattera and Purinton (2004)).

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of the potential role shopping malls could play in harnessing agglomeration effects almost as soon as cars came into wide-spread use and suburbs started being developed, it was not until changes in federal tax policy in the mid-1950s made the construction of new commercial real estate more attractive that they started to spread rapidly (Hanchett, 1996). As the share of the U.S. metropolitan population living in central cities went down from 69% in 1940 to 40% in 1990, businesses followed their customers and employees to suburbs and malls, large firms first, small firms later (Steinness, 1982). Perhaps as a response to these developments, the use of local development policies started growing rapidly in the 1970s (Andersson and Wassmer, 2000).[30] Based on our findings we would expect for this to lead to a decrease in the number of shopping malls opening up.[31] This is indeed what has happened, potentially setting the stage for the revival of bustling downtown leisure areas.

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[30] Appendix Figure 9 depicts the use of the term "development initiatives" in books published in the last 2000.

[31] We provide suggestive evidence of this effect in Appendix Table 20. Tax Increment Financing districts (TIFs) are atool used by local governments to subsidize retail development. As many as 46% of the cities pursuing retail development in the 2009 ICMA data report using this tool. TIFs can only be authorized after state-level enabling legislation is enacted, which happened at different times in different states. In the table, we regress the number of malls per capita opened for each state year on dummies for state, year, and whether TIF-enabling legislation has been enacted. Fewer malls opened after local governments had been authorized to use TIFs.

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Table 1. Exogeneity Check for Establishments: Pre-Trends

Dependent Variable	Δ ln Stores 2006-2007	Δ ln Employment 2006-2007	Δ ln Stores 2003-2007	Δ ln Employment 2003-2007
	(1)	(2)	(3)	(4)
0.5 Mile Radius				
Eventually Defunct Store	-0.00233	-0.0125	0.00139	-0.00477
	(0.00545)	(0.00759)	(0.00438)	(0.00477)
1 Mile Radius				
Eventually Defunct Store	-0.00229	-0.00518	0.00278	-0.00561
	(0.00341)	(0.00475)	(0.00366)	(0.00398)
2 Mile Radius				
Eventually Defunct Store	-0.00044	-0.00315	0.00409	-0.00485*
	(0.00217)	(0.00303)	(0.00294)	(0.00294)
3 Mile Radius				
Eventually Defunct Store	0.000853	-0.000962	0.00127	-0.00427
	(0.00178)	(0.00230)	(0.00316)	(0.00332)
N	7,003	7,003	6,174	5,355

Notes: The observations are outcomes for geographic neighborhoods of different sizes surrounding big box retailers. Each coefficient represents a separate regression. All retail chains were extant through 2007, with defunct stores closing between 2008 and 2012. A list of retailers can be found in the text. The reported coefficients are from a regression of the change in log stores and employment from 2006-2007 or 2003-2007 on a dummy for those chains that eventually close. The data for 2003 were provided in a different format. Due to concerns about data quality in some areas (we remove >40% swings), this generated different observation counts across columns. The table demonstrates the absence of pre-closure trend differences. Standard errors are clustered by zip code.

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Table 2. Spillovers from Big-Box Stores: Establishments and Employment

	0.25-Mile Radius		0.5-Mile Radius		1 Mile Radius		2 Mile Radius	
	ln Stores	ln Emp	ln Stores	ln Emp	ln Stores	ln Emp	ln Stores	ln Emp
Pooled	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	-0.117***	-0.131***	-0.0652***	-0.0798***	-0.0300***	-0.0314***	-0.0182***	-0.0105*
	(0.0134)	(0.0157)	(0.00884)	(0.0114)	(0.00472)	(0.00658)	(0.00320)	(0.00450)
Zip × Year								
Fixed Effects	-0.0819***	-0.0754***	-0.0374***	-0.0344***	-0.00805*	-0.0110*	-0.00196	-0.00553
	(0.0184)	(0.0209)	(0.0103)	(0.0123)	(0.00444)	(0.00577)	(0.00225)	(0.00352)
Only 1 Store per								
Type in Zip Code	-0.107***	-0.0953***	-0.0548***	-0.0594***	-0.0312***	-0.0284***	-0.0206***	-0.0167***
	(0.0208)	(0.0240)	(0.0140)	(0.0169)	(0.00701)	(0.00991)	(0.00442)	(0.00609)

Notes: This table reports the results of regressions of the form $\ln(y_{it}) = a_i + a_t + b \times \text{Defunct} + u_{it}$ where a_i are fixed effects for the neighborhood (or radius) and a_t are year (or year-store type in pooled specification) fixed effects. The observations are annual outcomes for neighborhoods of various sizes around big-box retailers from 2006-2012. All reported coefficients are for Defunct, which is a time-varying dummy equal to 1 if the big-box chain is closed in that year, and each coefficient represents a separate regression. Standard errors are clustered by zip code.

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Table 3. Spillovers from Big-Box Stores: Incumbency and Industry

	ln Stores			
Defunct - Non Defunct Pair	Foot Traffic	Non-Foot Traffic	Non-Competitor	Competitor
	(1)	(2)	(3)	(4)
Defunct	-0.0885***	-0.0109	-0.0321***	0.0418***
	(0.011)	(0.008)	(0.00414)	(0.00841)
N	48896	30052	55807	55807
Defunct - Non Defunct Pair	Entrant	Incumbent	Internet Penetration >Median	Internet Penetration <Median
	(5)	(6)	(7)	(8)
Defunct	-0.0307***	-0.0244***	-0.0429***	-0.0216***
	(0.00824)	(0.00428)	(0.0123)	(0.00732)
N	47834	47834	24822	25361

Notes: This table reports the results of regressions of the form $\ln(y_{it}) = a_i + a_t + b \times \text{Defunct} + u_{it}$ where a_i are fixed effects for the neighborhood (or radius) and a_t are year-store type fixed effects. The observations are annual outcomes for neighborhoods of half a mile around big-box retailers from 2006-2012. Each coefficient represents a separate regression. Defunct is a time-varying dummy equal to 1 if the big box chain is closed in that year. Businesses classified as NAICS 44 and 45 are counted as reliant on foot traffic, business classified as NAICS 11, 21, 22, 23, 31, 32, 33, 42, 48, 49, 51, 55, 56, 92, and 99 are not. Competitors are establishments in the same four digit NAICS code, otherwise they are a non-competitor. Entrant or incumbency status is determined annually for each establishment based on existance in the BA data the prior year. Internet penetration is computed at the state level using data from the Census. Standard errors are clustered by zip code.

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Table 4. Town Shape and Externalities

	ln Stores (Town)		ln Employment (Town)		ln Stores (1-Mile Radius)		ln Employment (1-Mile Radius)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Pooled								
Defunct	-0.00734***	-0.00775***	-0.00677**	-0.00576**	-0.0255***	-0.0272***	-0.0303***	-0.0303***
	(0.00253)	(0.00215)	(0.00279)	(0.00282)	(0.00482)	(0.00457)	(0.00589)	(0.00620)
Defunct X Geographic Ratio		-0.00931***		-0.00690**		-0.00650		-0.000462
		(0.00337)		(0.00315)		(0.00467)		(0.00576)
Measure	-	Compactness Index	-	Compactness Index	-	Compactness Index	-	Compactness Index
N	46358	46358	46358	46358	46357	46357	46355	46355

Notes: This table reports the results of regressions of the form $\ln(y_{it}) = a_i + a_t + b_1$ x Defunct + b_2 x Defunct x Geographic Ratio + u_{it} where a_i are fixed effects for the town or neighborhood, and the a_t are year fixed effects. The observations in Columns 1 through 4 are the annual outcomes for the town/political unit containing big-box retailers. The observations in Columns 5 through 8 are annual outcomes for 1-mile radii around big-box retailers. Defunct is a time-varying dummy equal to 1 if the big-box chain is closed in that year. The standard errors are clustered by zip code.

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Table 5. Compactness and Local Development Policy

Dependent Variable	Survey Retail Bidding Indicator				
	(1)	(2)	(3)	(4)	(5)
Compactness	0.0553***	0.0679***	0.0545***	0.0328**	0.0254*
	(0.0141)	(0.0129)	(0.0130)	(0.0155)	(0.0135)
N	2015	2015	1562	1993	2015

Notes: The table reports regressions of the form Retail Subsidies$_i$ = a + b x Compactness + cX$_i$ +u$_i$, where retail subsidies is an indicator for reporting retail development focus in the ICMA 1999, 2004, and 2009 Economic Development Surveys. Each coefficient represents a separate regression. The compactness measure is described in the text. All regressions include controls for area and year. Column 1 regressions include no additional control variables. Column 2 reports the coefficients controlling for population bin dummies, state dummies, log median house price, and the percentages of the population that do not speak English, are in poverty, and hold a bachelor's degree. Column 3 controls for the log dollars spent on development strategies and the total number of development strategies (retail and non-retail) pursued by the responding town. Column 4 regressions control for the responding town's indication that they have a large retail tax base and a sales tax. Column 5 reports the coefficients controlling for metro (CSA) fixed effects.

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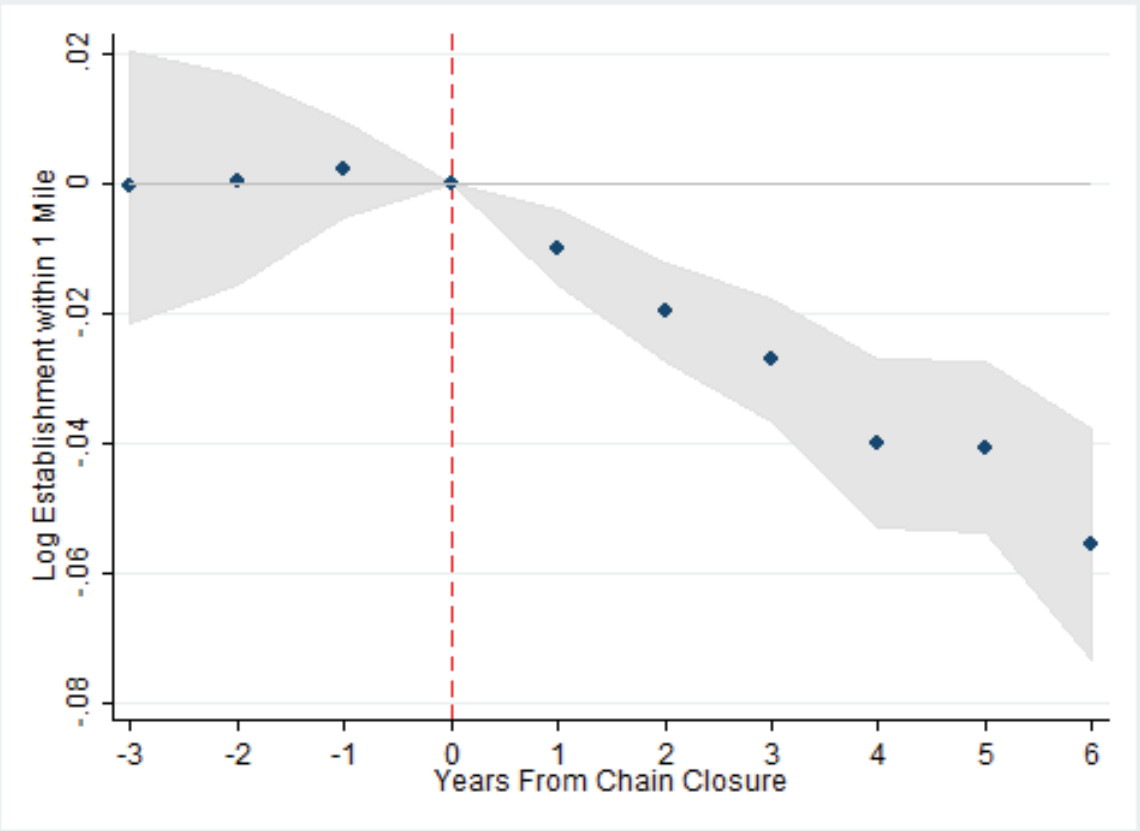
Table 6. Compactness, Shopping Malls and Business Improvement Districts

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Shopping Center Indicator			Log (Shopping	Log (Center	BID
	All Distances		$<$ 2 Miles	$>$ 2 Miles	Centers)	Stores)	Indicator
			from Border	from Border			
Retail Focus	-0.0487*						
	(0.0262)						
Compactness		-0.0472***	-0.0449***	-.0010104	-0.0525**	-0.0661**	-0.114**
		(0.00568)	(.0054435)	(.0008895)	(0.0208)	(0.0311)	(0.0348)
N	1948	28940	28940	28940	2219	2217	293

Note: The table reports regressions of the form Y_i = a + b x Compactness + cX_i +u_i where the dependent variable is (in order) a dummy for the existence of a mall or shopping center, the log number of malls/centers, the log number of stores in malls/centers, and the existence of a business improvement district. The explanatory variable is our compactness index, except for in Column 1, where the explanatory variable is a dummy for retail focus in the ICMA surveys. Each coefficient represents a separate regression and all regressions control for area. Standard errors clustered by state in parentheses.

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Figure 1. Event Study

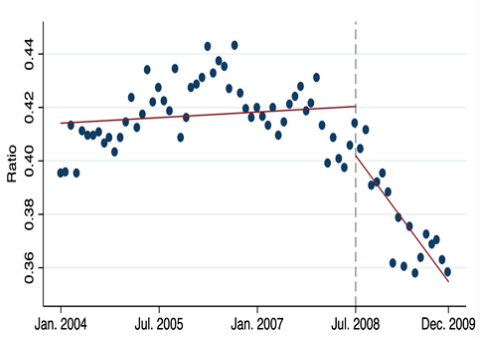


Notes: This figure displays the coefficients b_t and confidence interval from the regression: $\ln(y_{it})=a_i+a_t$ x 1(t=T)+ b_t x Defunct x 1(t=T) $+u_{it}$ where a_i are fixed effects for the neighborhood (or radius) and a_t are year fixed effects. The observations are annual outcomes for neighborhoods around big-box retailers from 2006-2012. Defunct is a time-varying dummy equal to 1 if the big-box chain is closed in that year. Standard errors are clustered by zip code.

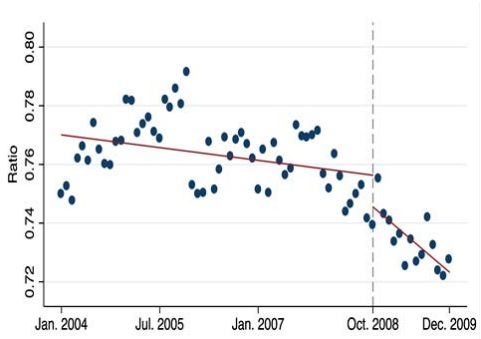
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Figure 2. Spillovers from Big-Box Stores: Consumer Behavior

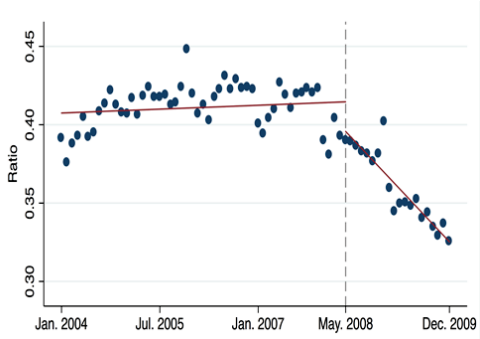
Aggregate Ratios



Electronics: Ratio of Same-Day Visits to Total Visits

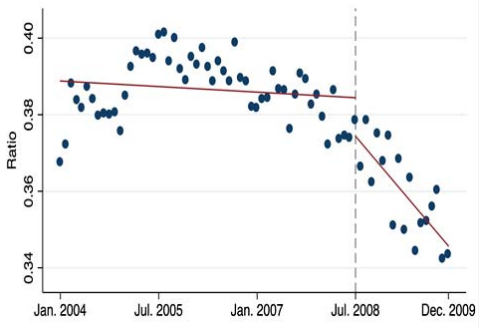


Department: Ratio of Same-Day Visits to Total Visits

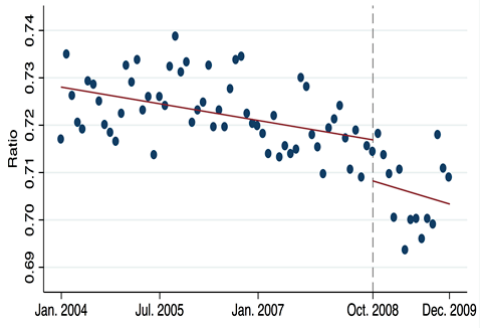


Furnishings: Ratio of Same-Day Visits to Total Visits

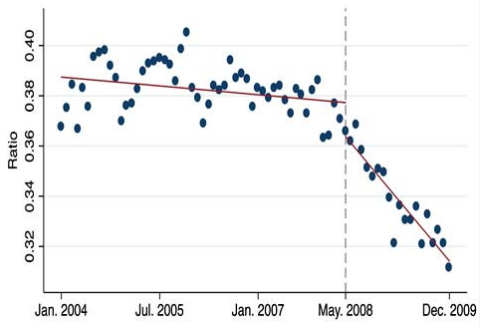
Individual Fixed Effects



Electronics: Ratio of Same-Day Visits to Total Visits



Department: Ratio of Same-Day Visits to Total Visits



Furnishings: Ratio of Same-Day Visits to Total Visits

Notes: These graphs depict the ratio of visits at stores visited on the same day as the imputed big-box store to total visits at all stores. Observations are binned at the year-month level and are restricted to households observed in 48 or more of the year-month bins in defunct zip codes only. Closing date is May 2008 for Linens 'n Things, July 2008 for Circuit City and CompUSA, and October 2008 for Mervyn's.